3.4.4 Some Characteristics of Trigonometric Functions

Trigonometric functions have some nice properties. For example,

(i) Sine and cosine functions are complementary to each other in the sense that $\sin(90^\circ - \theta) = \cos\theta$ and $\cos(90^\circ - \theta) = \sin\theta$.

(ii) As $\cos\theta$ and $\sin\theta$ are obtained as coordinates of a point on the unit circle, they satisfy the inequalities $-1 \leq \cos\theta \leq 1$ and $-1 \leq \sin\theta \leq 1$. Hence, $\cos\theta, \sin\theta \in [-1, 1]$

(iii) Trigonometric function repeats its values in regular intervals.

(iv) Sine and cosine functions have an interesting property that $\cos(-\theta) = \cos\theta$ and $\sin(-\theta) = -\sin\theta$

Let us discuss the last two properties.

Periodicity of Trigonometric Functions

We know that a function f is said to be a periodic function with period p, if there exists a smallest positive number p such that $f(x + p) = f(x)$ for all x in the domain.
For example, $\sin(x + 2n\pi) = \sin x, n \in \mathbb{Z}$.

i.e., $\sin(x + 2\pi) = \sin(x + 4\pi) = \sin(x + 6\pi) = \ldots = \sin x$

Thus, $\sin x$ is a periodic function with period 2π.
Similarly, $\cos x$, cosec$x$ and $\sec x$ are periodic functions with period 2π.

But $\tan x$ and $\cot x$ are periodic functions with period π.

The periodicity of $\sin x$ and $\cos x$ can be viewed best using their graphs.

(i) **The graph of the sine function**

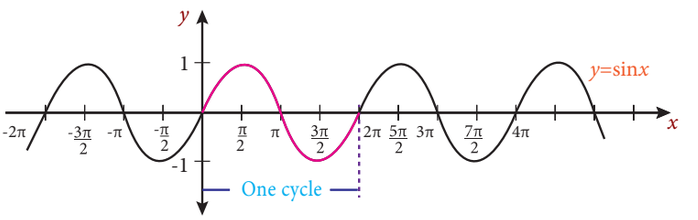


Figure 3.13: y =sinx

Here x represents a variable angle. Take the horizontal axis to be the x-axis and vertical axis to be the y-axis. Graph of the function $y = \sin x$ is shown in the Figure 3.13. First, note that it is periodic of period 2π. Geometrically it means that if you take the curve and slide it 2π either left or right, then the curve falls back on itself. Second, note that the graph is within one unit of the y-axis. The graph increases and decreases periodically. For instance, increases from $-\frac{\pi}{2}$ to $\frac{\pi}{2}$ and decreases from $\frac{\pi}{2}$ to $\frac{3\pi}{2}$.

(ii) **The graph of the cosine function**

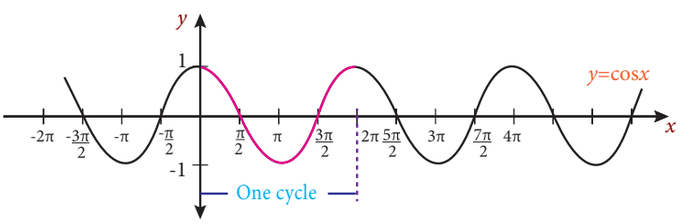


Figure 3.14: y = cosx